Student ID:		Lab Section:	
Name:		Lab Group:	

Experiment No. 6

Verification of Thevenin's Theorem and Maximum Power Transfer Theorem

Objective

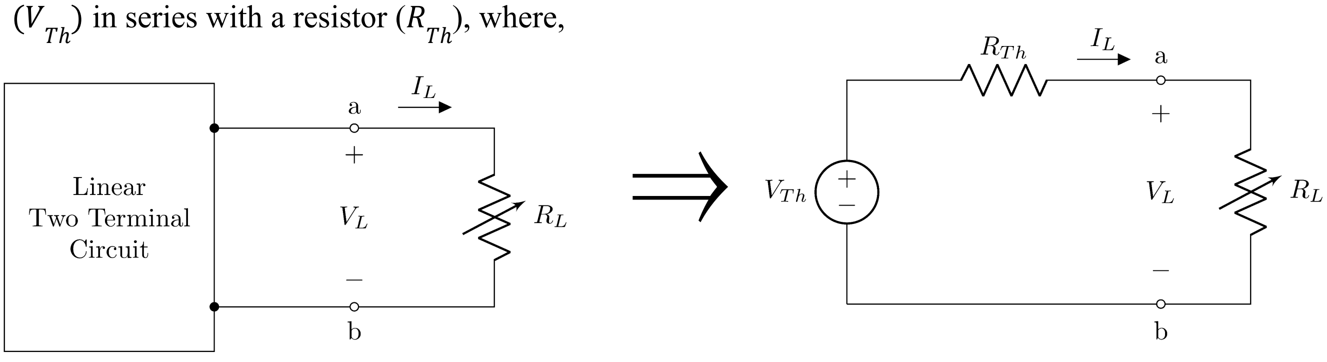
This experiment aims to validate Thevenin's Theorem for linear circuits as well as the condition for the Maximum Power to be delivered to the load of any linear two-terminal circuit.

Part 1: Thevenin's Theorem

Theory

It is often desirable in circuit analysis to study the effect of changing a particular branch element while all other branches and all the sources in the circuit remain unchanged. Thevenin's theorem is a technique to this end, and it greatly reduces the number of computations that we have to do each time a change is made. Using Thevenin's theorem the given circuit except for the particular branch to be studied is reduced to the simplest equivalent circuit possible and then the branch to be changed is connected across the equivalent circuit.

Thevenin's theorem states that any two-terminal linear bilateral networks containing sources and passive elements can be replaced by an equivalent circuit consisting of a voltage source (V_{Th}) in series with a resistor (R_{Th}), where,



V_{Th} = The open circuit voltage (V_{OC}) at the two terminals a and b.

R_{Th} = The resistance looking into the terminals a and b of the network with all sources removed.

There are several methods for determining Thevenin resistance R_{Th}. An attractive method for determining R_{Th} is: (1) determine the open circuit voltage, and (2) determine the short circuit current I_{SC} as shown in the figure; then

Methodology of Determining Thevenin's Circuit Parameters (V_{OC}, I_{SC}, R_{th})

The procedure to determine Thevenin's Circuit Parameters (V_{OC}, I_{SC}, R_{Th}) and Thevenin equivalent circuit for any linear two-terminal linear circuit is given below.

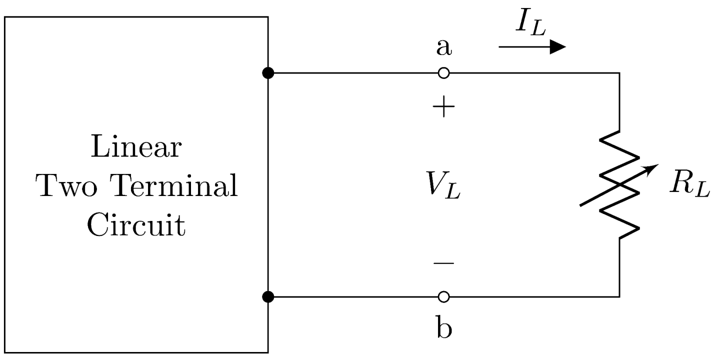


Figure 1: Original Circuit

Step 1: Determining V_{OC}

Remove the load resistance R_L and find the open circuit voltage between terminals a & b. This voltage is called Thevenin's voltage, i.e., $V_{Th} = V_{OC}$, also known as the Open Circuit Voltage.

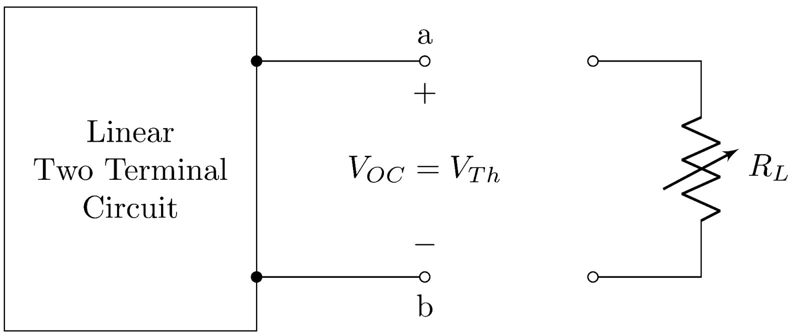


Figure 2: Circuit for finding V_{OC}

Step 2: Determining I_{SC}

Place a short circuit between terminals a and b (simply connect them through a wire). The current through the short circuit is called Norton's Current, i.e., $I_N = I_{SC}$. It is also known as the Short Circuit Current.

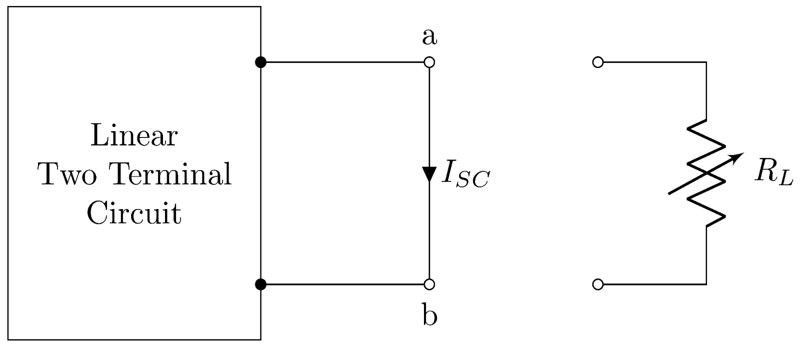


Figure 3: Circuit for finding I_{SC}

Step 3: Determining R_{th}

Divide the Open Circuit Voltage ($V_{OC} = V_{Th}$) by the Short Circuit Current (I_{SC}) to determine the Thevenin's Resistance.

$$R_{Th} = \frac{V_{OC}}{I_{SC}}$$

Alternatively, turn off all the independent sources and determine the equivalent resistance between terminals $a - b$. This is the Thevenin resistance, that is, $R_{ab} = R_{Th}$.

Step 4: Constructing Thevenin's Equivalent Circuit

Construct Thevenin's equivalent circuit as shown in the following figure setting the voltage source at V_{Th} volts and the series resistance at R_{Th} ohms. The values shown here should closely match the values you determined from the earlier steps.

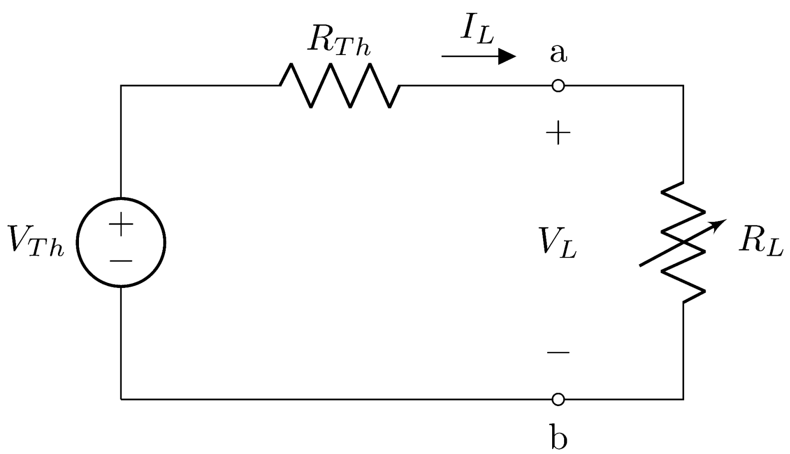


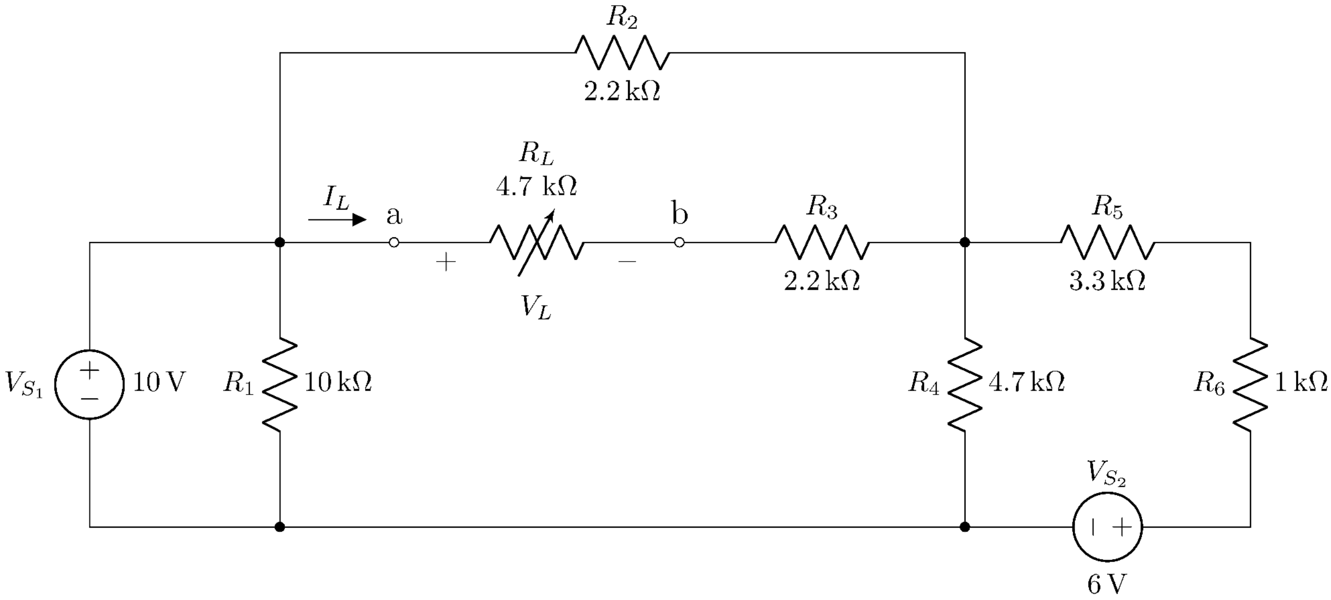
Figure 4: Thevenin Equivalent Circuit

Apparatus

- ➢ Multimeter
- ➢ Resistors ($1\ k\Omega,\ 2.2\ k\Omega$ x $2,\ 3.3\ k\Omega,\ 4.7\ k\Omega$ x $2,\ 10\ k\Omega$).
- ➢ DC power supply
- ➢ Breadboard
- ➢ Jumper wires

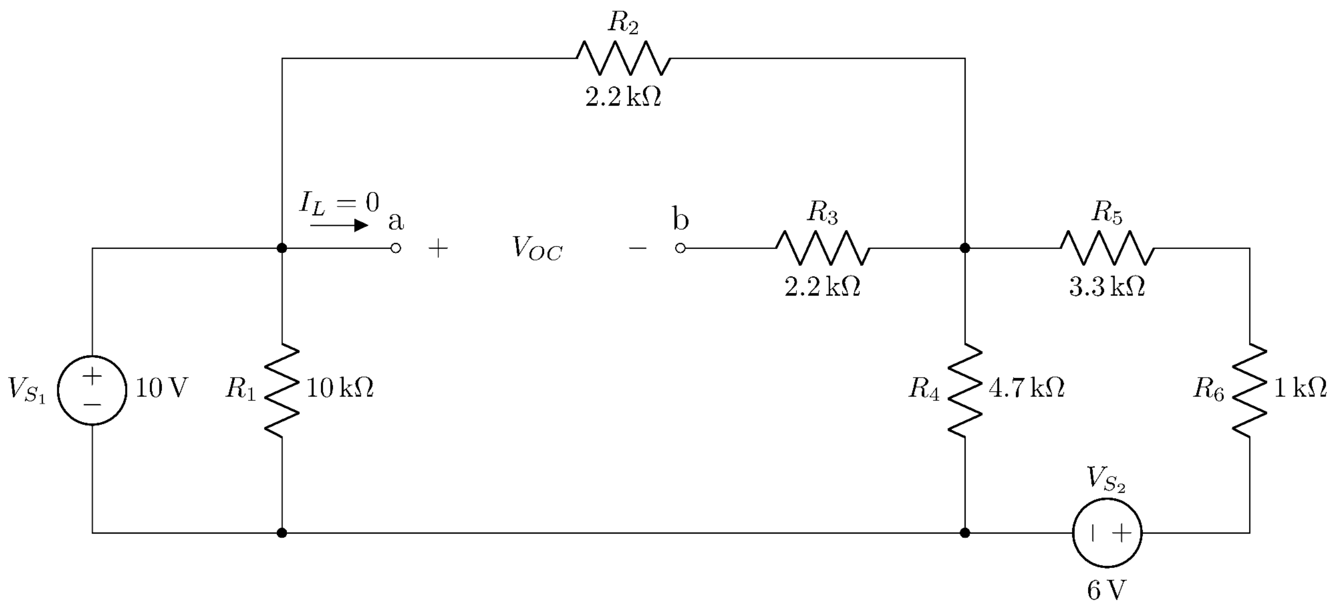
Procedures

- ➢ Measure the resistances of the provided resistors and fill up the data table.
- ➢ Construct the following circuit on a breadboard. Try to use as less wires as possible.



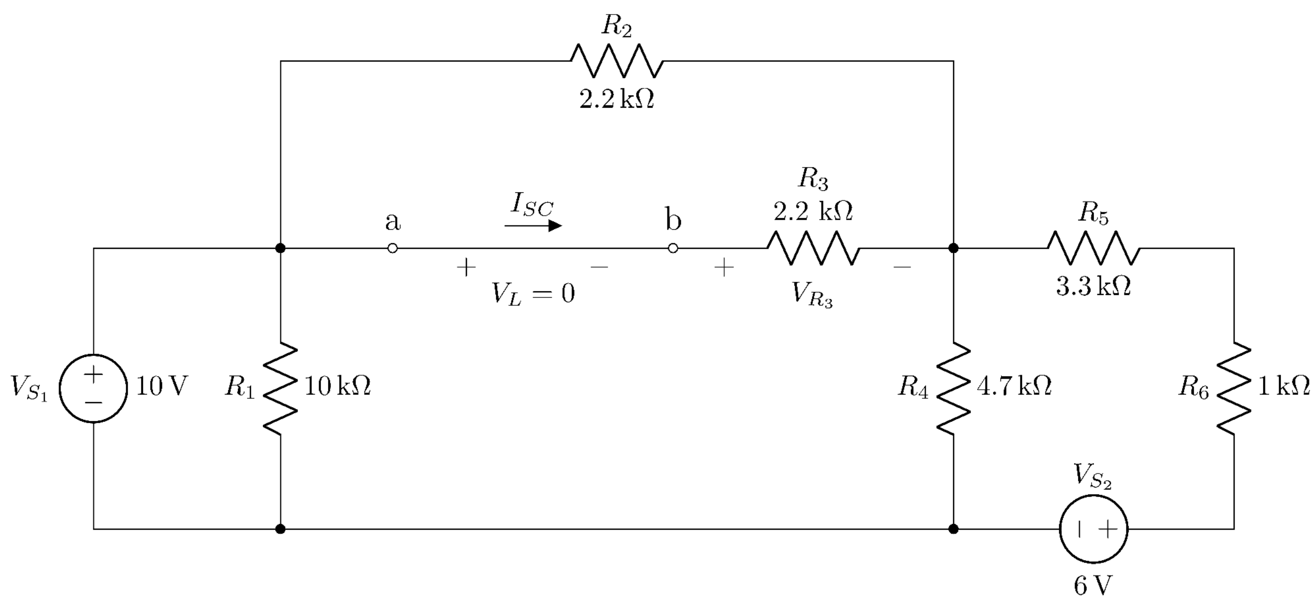
Circuit 1

- ➢ Connect two DC voltage sources or two channels of a DC source with voltages set to $6\ V$ and $10\ V$ as shown in the figure.
- ➢ We will model the load resistance (R_L) by a $4.7\ k\Omega$ resistor for simplicity. Connect a $4.7\ k\Omega$ resistor as R_L.
- ➢ Measure the voltage (V_L) across the load resistance. Measure the current (I_L) through the load using Ohm's law as $I_L = \frac{V_L}{R_L}$. Record the values in the corresponding data table.
- ➢ Now, disconnect the load resistor and leave the terminals $a - b$ open. The circuit should look like the one shown below.



Circuit 2

- Notice that, as the load is open-circuited, the load current (I_L) is zero. Measure the open circuit voltage V_{OC} and record it in the corresponding data table.
- Now short the terminals $a - b$ by connecting a wire between them as shown in the following diagram.



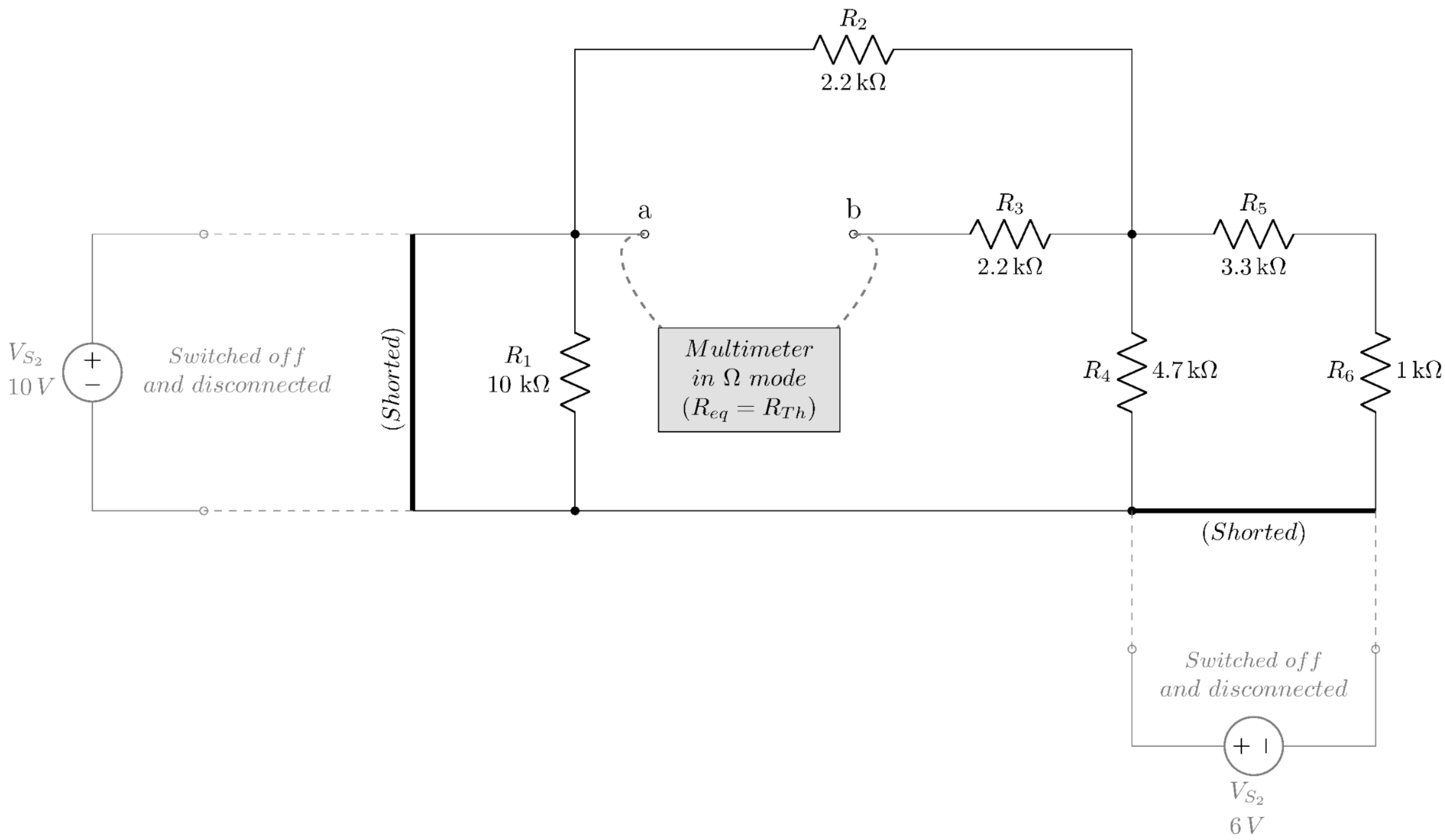
Circuit 3

- Notice that, as the load is short-circuited, the load voltage (V_L) is zero. Measure the short circuit current I_{SC} and record it in the corresponding data table. I_{SC} is the current flowing through the R_3 resistor, that is $I_{SC} = \frac{V_{R_3}}{R_3}$.

➢ Use V_{Th} from **Circuit 2** and I_{SC} from **Circuit 3** to calculate R_{Th} as $R_{Th} = \frac{V_{Th}}{I_{SC}}$ in data Table 5.

➢ Now let's determine R_{Th} using another method which is called the Universal Method. To do this, again open the terminals $a - b$ (remove the shorting wire in **Circuit 3**).

➢ Switch off the voltage source(s). Replace them with short circuits as shown in the following diagram.

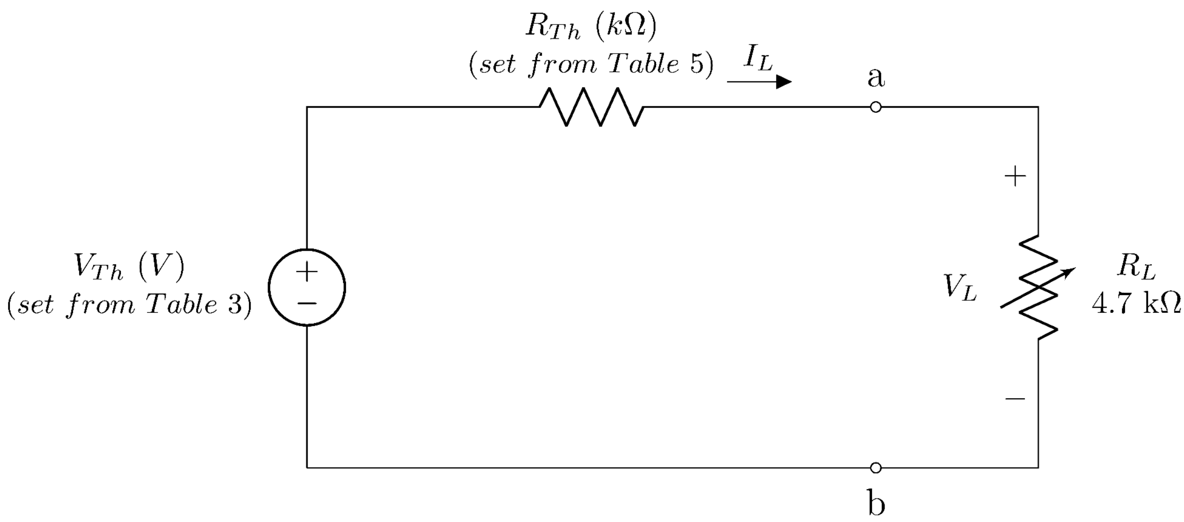


Circuit 4

➢ Measure the equivalent resistance between terminals $a - b$. This the Thevenin resistance R_{Th}.

➢ Now construct the following Thevenin equivalent circuit shown in the following diagram.

➢ Set the value of the voltage source equal to V_{Th} (from Table 3). Use R_{Th} from Table 5.



Circuit 5

- ➢ Measure the voltage (V_L) across the load resistance. Measure the current (I_L) through it using Ohm's law as $I_L = \frac{V_L}{R_L}$. Record the values in the corresponding data table.

Data Tables

Signature of Lab Faculty: | **Date:**

** ***For all the data tables, take data up to three decimal places, round to two, then enter into the table.***

Table 1: Resistance Data

For all your future calculations, please use the observed values only **(even for theoretical calculations)**.

Notation	Expected Resistance	Observed Resistance (kΩ)	Notation	Expected Resistance	Observed Resistance (kΩ)
R_1	10 kΩ		R_5	3.3 kΩ	
R_2	2.2 kΩ		R_6	1 kΩ	
R_3	2.2 kΩ		R_L	4.7 kΩ	
R_4	4.7 kΩ				

Table 2: Data from Circuit 1

In the following table, V_L is the voltage drop across the load resistor R_L and I_L is the current through the load with polarity and direction respectively as shown in **Circuit 1**. Solve the circuit and calculate theoretical V_L and I_L. Also, calculate the percentage of error between experimental and theoretical values of V_L.

Observation	V_{S_1} (V) (from dc supply)	V_{S_1} (V) (using multimeter)	V_{S_2} (V) (from dc supply)	V_{S_2} (V) (using multimeter)	V_L (V)	$I_L = \frac{V_L}{R_L}$ (mA)
Experimental						
Theoretical						

$$\textbf{Percentage of Error} = \left|\frac{Experimental - Theoretical}{Theoretical}\right| \times 100\%$$

Here, Percentage of Error in V_L calculation = %

Table 3: Data from Circuit 2

In the following table, V_{OC} is the open circuit voltage across the open terminals with $I_L = 0$. This is the Thevenin voltage V_{Th}. Calculate the percentage of error between experimental and theoretical values of V_{OC}.

Observation	V_{S_1} (V) (from dc supply)	V_{S_1} (V) (using multimeter)	V_{S_2} (V) (from dc supply)	V_{S_2} (V) (using multimeter)	$V_{OC} = V_{Th}$ (V)
Experimental					
Theoretical					

Here, % error in $V_{OC} = V_{Th}$ calculation = %

Table 4: Data from Circuit 3

In the following table, I_{SC} is the current through the shorted terminals with $V_L = 0$. Theoretically calculate the short circuit current and calculate the percentage of error between experimental and theoretical values of I_{SC}.

Observation	V_{S_1} (V) (from dc supply)	V_{S_1} (V) (using multimeter)	V_{S_2} (V) (from dc supply)	V_{S_2} (V) (using multimeter)	V_{R_3} (V)	$I_{SC} = \frac{V_{R_3}}{R_3}$ (mA)
Experimental						
Theoretical						

Here, % error in I_{SC} calculation = %

Table 5: R_{Th} calculation

Comparison of the two methods to determine R_{Th}.

Here, $\Delta R_{Th} = (R_{Th}\ using\ V_{Th}/I_{SC}\ method) - (R_{Th}\ using\ the\ Universal\ method)$

Observation	V_{Th} (V) (from data Table 3)	I_{SC} (mA) (from data Table 4)	$R_{Th} = \frac{V_{Th}}{I_{SC}}$ (kΩ)	R_{Th} (kΩ) (using multimeter from Circuit 4)	ΔR_{Th} (kΩ)
Experimental					
Theoretical					

Table 6: Data from Circuit 5

In the following table, V_L is the voltage drop across the load resistor R_L and I_L is the current through the load with polarity and direction respectively as shown in **Circuit 1** and **Circuit 5**. Measure the value of V_L from **Circuit 5**. Then calculate I_L using the measured V_L. Finally, compare the values with those in Table 1. Here,

$\Delta V_L = (V_L\ from\ Table\ 2) - (V_L\ measured\ for\ Circuit\ 5$ in Table 6) and

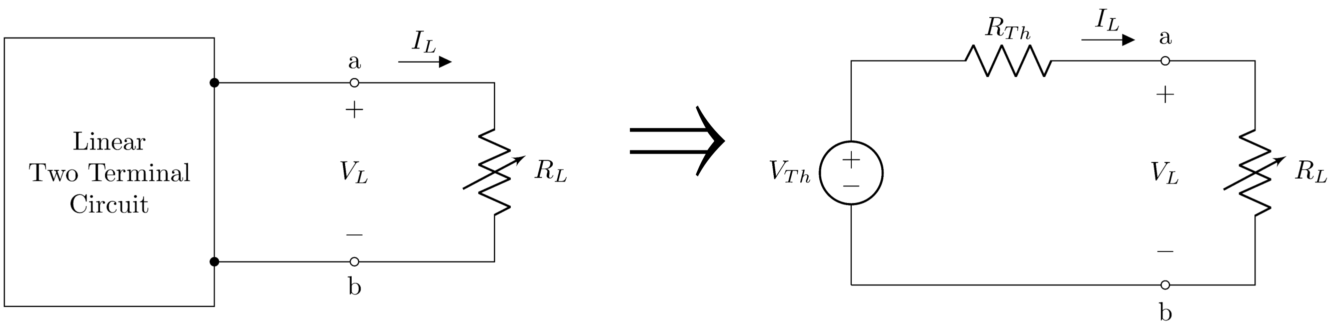
$\Delta I_L = (I_L\ from\ Table\ 2) - (I_L\ calculated\ for\ Circuit\ 5$ in Table 6).

Observation	V_L (V) (from data Table 2)	I_L (mA) (from data Table 2)	V_L (V) (from Circuit 5 using multimeter)	$I_L = \frac{V_L}{R_L}$ (V) (for Circuit 5)	ΔV_L (V)	ΔI_L (mA)
Experimental						
Theoretical						

Part 2: Maximum Power Transfer Theorem

Theory

The Maximum Power Transfer Theorem is a fundamental concept in electrical engineering that relates to the transfer of maximum power from a source to a load. The Maximum Power Transfer theorem states that ***A resistive load will receive maximum power when its total resistive value is exactly equal to Thevenin's resistance of the network as "seen" by the load.***



We know that any circuit A terminated with a load R_L can be reduced to its Thevenin equivalent. Now according to this theorem, the load R_L will receive maximum power when $R_L = R_{Th}$. We can calculate the Maximum Power theoretically using the formula,

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

The theorem focuses on the transfer of power between a source and a load. In electrical circuits, power is transferred from a source (such as a generator) to a load (such as a resistor) through a transmission medium (such as wires or conductors).

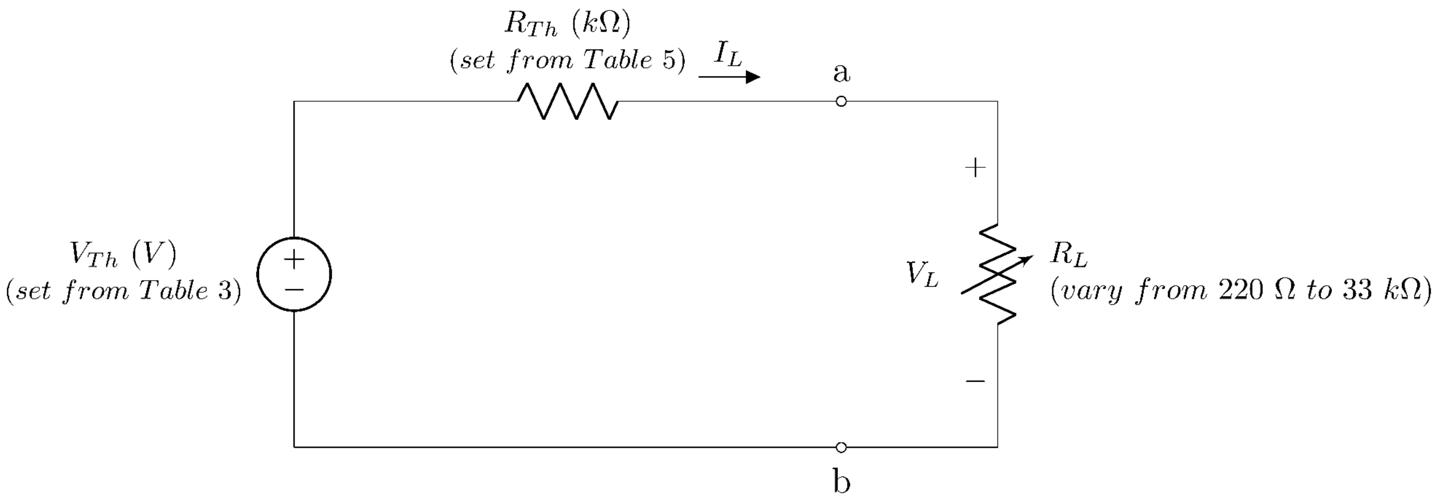
It's worth noting that the Maximum Power Transfer Theorem is a theoretical concept and is not always practical or desirable in real-world scenarios. In many practical applications, impedance matching is employed to achieve efficient power transfer, but it may not always result in maximum power transfer. Design considerations, system constraints, and other factors often influence the choice of impedance matching in electrical circuits.

Apparatus

- Multimeter
- Resistors ($220\ \Omega,\ 1\ k\Omega,\ 1.5\ k\Omega,\ 2.2\ k\Omega,\ 3.3\ k\Omega$ x 2, $4.7\ k\Omega,\ 5.6\ k\Omega,\ 10\ k\Omega,$ $18\ k\Omega$, $33\ k\Omega$) or ($20\ k\Omega$ Potentiometer and $33\ k\Omega$).
- DC power supply
- Breadboard
- Jumper wires

Procedures

- Construct the following reduced Thevenin equivalent circuit.



Circuit 6

- Vary R_L from 220 Ω to 33 kΩ. For each resistor listed in the **Apparatus** section, measure V_L. Calculate I_L using $I_L = \frac{V_L}{R_L}$ and record in the corresponding data table.

Data Tables

Signature of Lab Faculty:		**Date:**	

**** *<u>For all the data tables, take data up to three decimal places, round to two, then enter into the table.</u>***

Table 7: Data from Circuit 6

In the following table, P_{In} is the power supplied by the dc source of whose value is set equal to V_{Th}, and P_{Load} is the power consumed by the load. η is the power efficiency. Theoretically, η is 50% at the maximum power transfer condition.

R_L (Expected)	R_L (kΩ) (Measured)	V_{Th} (V) (using multimeter)	V_L (V)	$I_L = \frac{V_L}{R_L}$ (mA)	$P_{In} = V_{Th} I_L$ (mW)	$P_{Load} = V_L I_L$ (mW)	Efficiency $\eta = \frac{P_{Load}}{P_{In}} \times 100$ (%)
0.22 kΩ							
1 kΩ							
1.5 kΩ							
2.2 kΩ							
3.3 kΩ							
4.7 kΩ							
5.6 kΩ							
10 kΩ							
18 kΩ							
33 kΩ							

Maximum Power found from the Table 7, P_{max} = ______ mW

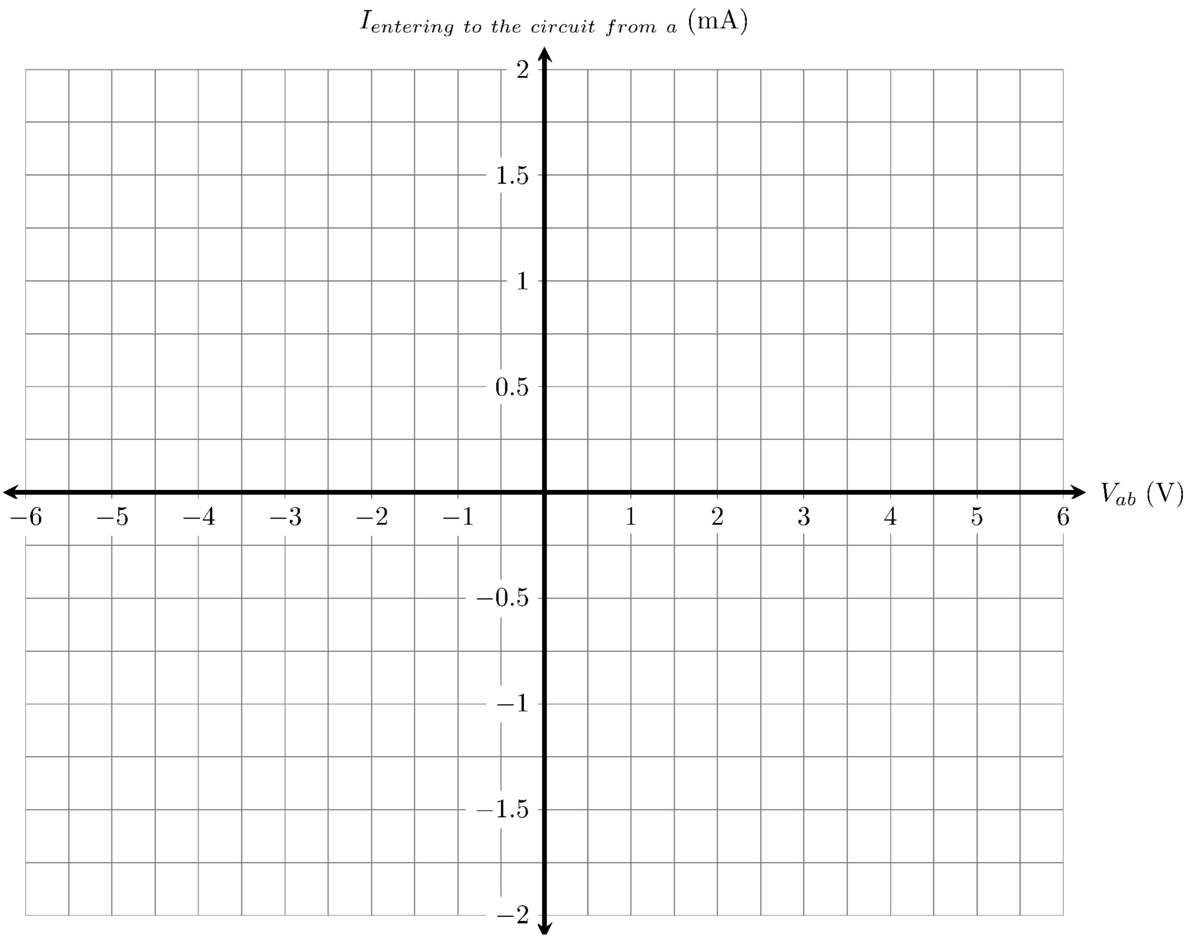
Theoretical Maximum Power, $P_{max} = \frac{V_{Th}^2}{4R_{Th}} =$ ______ mW

Here, percentage error in Maximum Powers calculation = ______ %

Questions

1. Circuit equivalency:

(a) Draw the $I - V$ characteristic of the **Circuit 5** to the left of terminals $a - b$ in the template provided below.



The straight line intersects x-axis at [V]

The straight line intersects y-axis at [mA]

Slope of the straight line, m = [$k\Omega^{-1}$]

Resistance from the plot, $\frac{1}{m}$ = [$k\Omega$]

(b) Comparing the values to V_{OC}, I_{SC}, and R_{Th} measured for **Circuit 1** in Tables 3, 4, and 5 respectively, do they closely match?

☐ Yes ☐ No

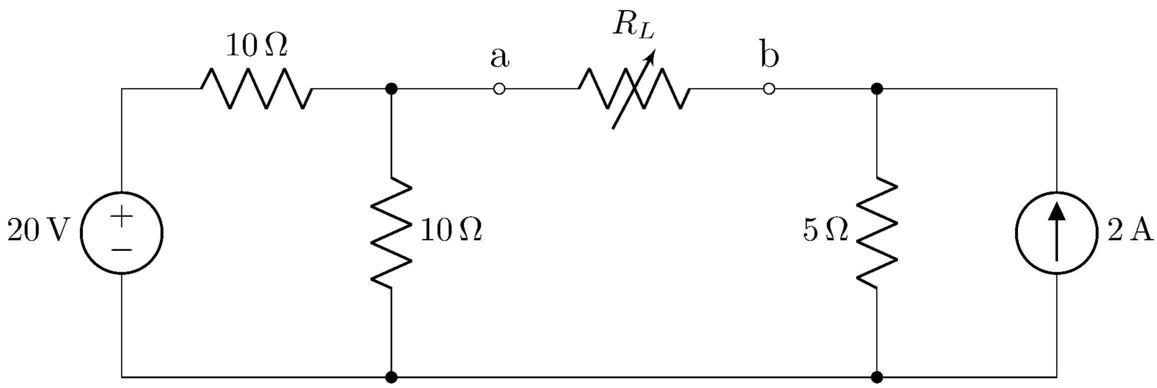
2. Now in the same plot provided above, plot the $I - V$ curve of **Circuit 4**.

Slope of the straight line, $m =$ [ ] $k\Omega^{-1}$

Resistance from the plot, $\frac{1}{m} =$ [ ] $k\Omega$

Explain why there is a shift in the $I - V$ curve of Circuit 4.

3. Determine the open circuit voltage (V_{OC}) and the short circuit current (I_{SC}) to the terminals $a - b$ for the circuit shown below.



Open circuit voltage, $V_{OC} =$ [ ] V

Short circuit current, $I_{SC} =$ [ ] mA

Have you been able to calculate the Thevenin resistance (R_{Th}) using $R_{Th} = \frac{V_{OC} = V_{Th}}{I_{SC}}$?

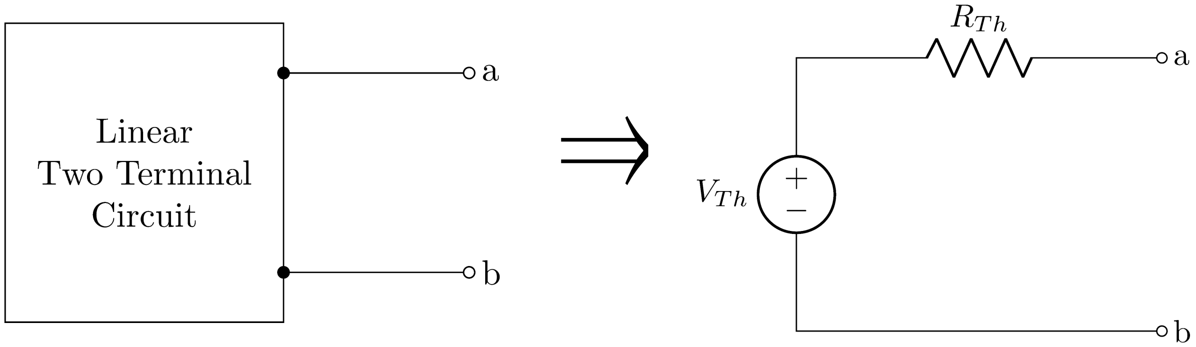
☐ Yes ☐ No

If not, suggest an alternative approach and determine R_{Th}.

We can determine R_{Th} by

Calculation:

4. If the following voltage source is in series with a resistor is the Thevenin equivalent of the linear two-terminal circuit, for each of the circuit elements listed in column 1 of the following table, write the values of V_{Th} and R_{Th}. Write '*Unknown*' if unable to specify.



The linear two-terminal circuit is composed of only a/an	V_{Th} (V)	R_{Th} (Ω)
Short circuit		
Open circuit		
$-$ 2 V ideal voltage source		
3 A ideal current source		
5 kΩ resistor		

5. Efficiency and Maximum Power:

(a) From the η and P_{Load} vs. R_L plot, what is the efficiency at the maximum power position?

η = % at the maximum power point position.

(b) For a load resistance R_L,

we can **increase** the power efficiency of the load by −

☐ Increasing R_L than R_{Th} ☐ Decreasing R_L than R_{Th} ☐ By equating R_{Th} and R_L

(c) We can **maximize** the power transfer of the load by −

☐ Increasing R_L than R_{Th} ☐ Decreasing R_L than R_{Th} ☐ By equating R_{Th} and R_L

(d) "We cannot maximize both the power of a load and the power efficiency of the circuit" − justify the statement.

6. Specify by putting × *or* ✔, what should be the first priority: maximizing the power or increasing the efficiency for the following applications −

Application	Should maximize the power transferred to the load	Should try to operate the load with the highest efficiency possible
An antenna sending a signal to Mars		
A motor running to pull water to a tank placed in a higher position		
A mic used to amplify a voice		

Report

1. Fill up the theoretical parts of all the data tables.
2. Answers to the questions.
3. Attach two data plots; one should include V_L, I_L, and P_{Load} vs. R_L plotted together in the same pane, and the other should include η and P_{Load} vs. R_L plotted together in the same pane. There is a guideline for plotting data using Google Sheets on the next page.
4. Discussion [brief summary of the experiment, key concepts learned, accuracy and consistency of the measured data with theory, difficulties experienced and your thoughts on those, and ways to improve]. Add pages if necessary.

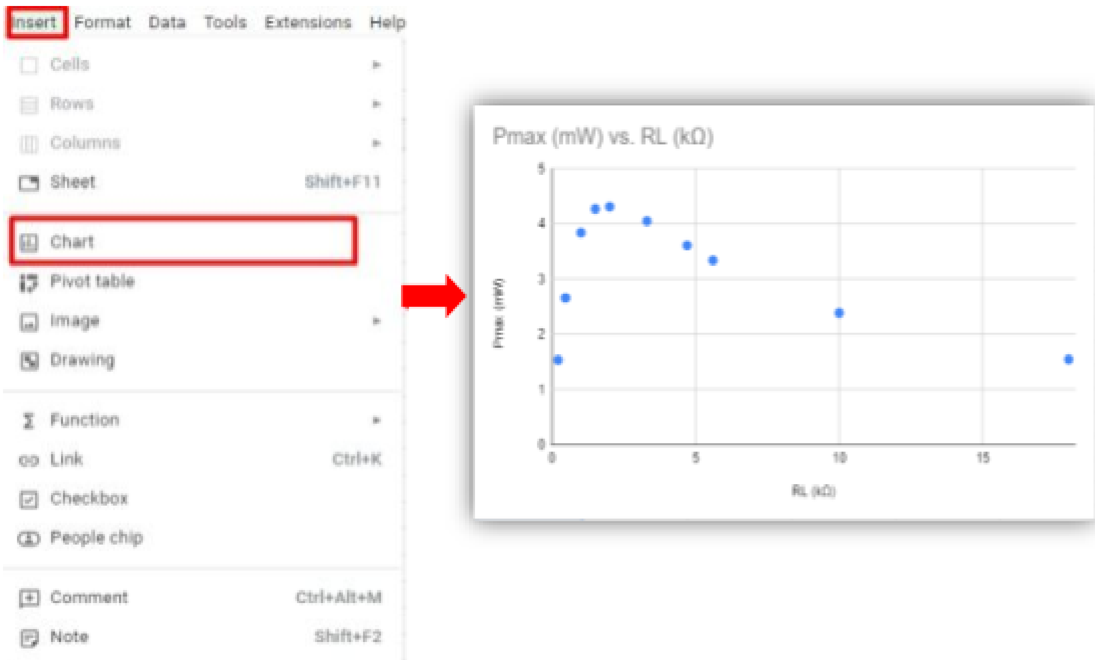
<u>Part 3: Plotting Circuit Characteristics on Google Sheets</u>

1. Create a Google spreadsheet by visiting https://docs.google.com/spreadsheets
2. Fill in the spreadsheet with the data that you've collected in the lab (refer to your lab sheet). Select the column R_L $(k\Omega)$ and any other column you want to plot with (to select a column, click on the column head, e.g., "A". Then hold CTRL while clicking the second column, e.g., "B", to select both columns.)

A	B	C	D	E	F	G	H
RL (kΩ)	Pmax (mW)	% η	% VR	Loss	Pout	IL	VL
0.22	1.54						
0.47	2.66						
1	3.84						
1.5	4.27						
2	4.31						
3.3	4.05						
4.7	3.61						
5.6	3.34						
10	2.39						
18	1.55						

Note: This is sample data collected from a simulation. Your data may not match with this.

3. Select Insert → Chart. You should be getting a graph that looks like the following diagram.



4. A Chart Editor section should pop up on the right side of your screen. If it doesn't show up, then double-click on the graph. Go to the setup section in the chart editor and change the "Chart type" to "Line chart". Your graph should be changed into a line plot as shown below.